

ARTIFICIAL INTELLIGENCE & NATURAL LANGUAGE PROCESSING (LECTURE 11-1)

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Introduction to Biomedical Informatics 1

Apr 28, 2025



LEARNING OBJECTIVES

- Define artificial intelligence
- Learn the main types and classes of AI
- Learn about various AI applications, particularly in BMS
- Learn the main approaches of natural language processing
- Learn about NLP applications, particularly in BMS
- Understand social and ethical considerations of AI



WHAT IS ARTIFICIAL INTELLIGENCE?

A bit like :

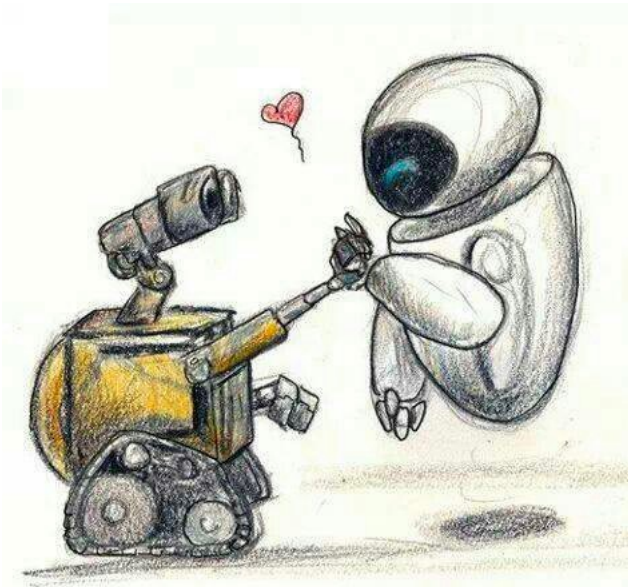
"Probability Symphony": Metaphor for the Essence of Technology ;

"Dynamic Mirror": Philosophical Reflection on the Relationship between Man and Machine ;

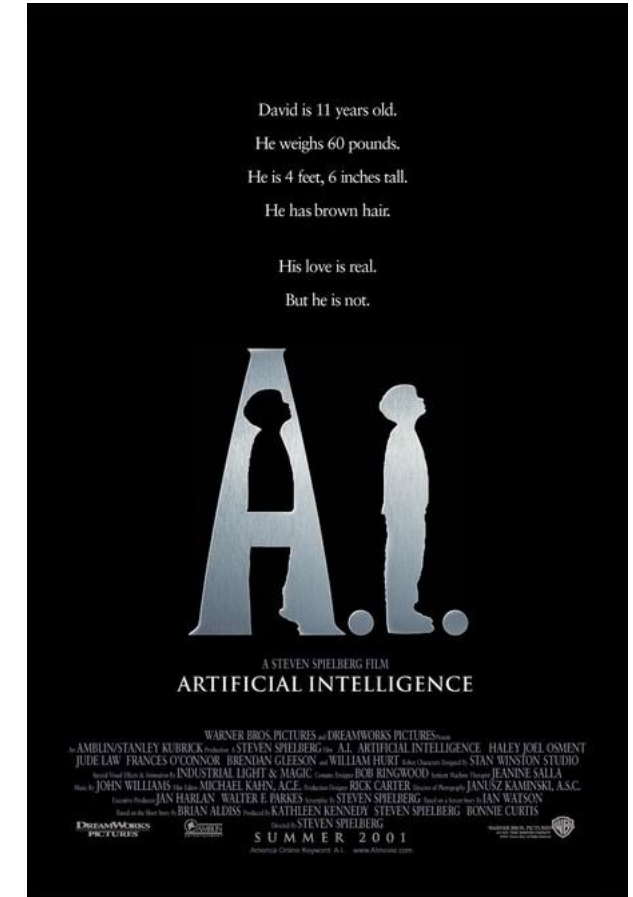
... ..



2001: A Space Odyssey



Wall-E



AI: A THEORETICAL DEFINITION

- A branch of computer science/informatics that aims to create intelligent machines

What does it mean “intelligent”?

- Able to perceive (see, hear, smell)
- Able to learn
- Able to make decisions
- Able to solve problems

Well, yes, but not necessarily...

HAL9000: A Heuristically programmed Algorithmic computer

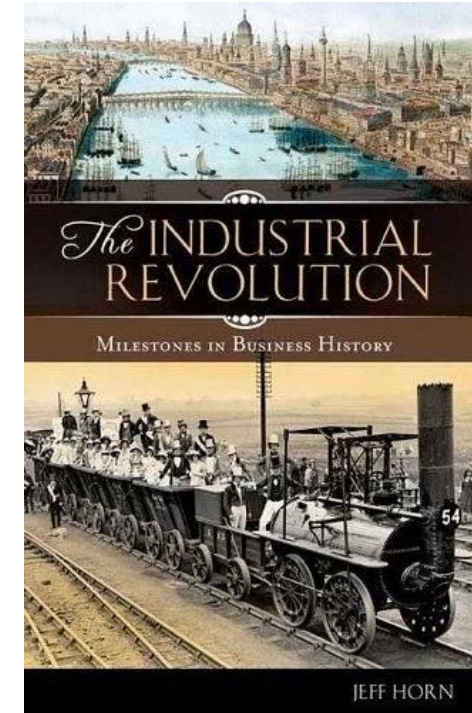
So basically, the HAL 9000 / Wall-E / David kind, right?



AI: A FOCUSED APPROACH

It's a lot harder to build a machine able to do all of them in a coordinated way than it is to focus on these tasks separately, e.g.

- **Problem solving** – various algorithms: text mining / natural language processing, game playing, statistical inference
- **Perception** – computer vision, speech processing/synthesis
- **Machine learning** – pattern discovery, classification, other prediction
- Combination of the three

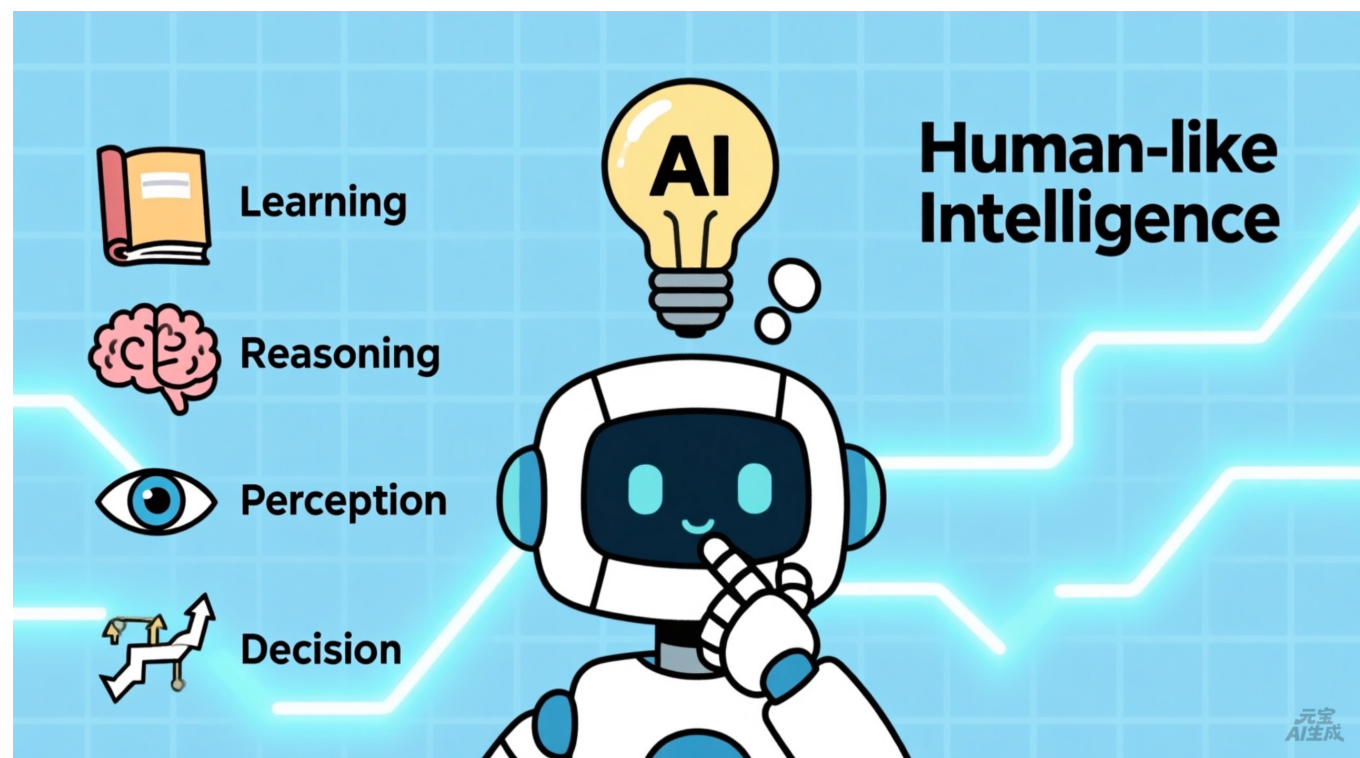


https://www.ted.com/talks/kevin_kelly_how_ai_can_bring_on_a_second_industrial_revolution



WHAT IS ARTIFICIAL INTELLIGENCE?

Artificial intelligence (AI) is the science and technology that enables machines to simulate human intelligence (such as learning, reasoning, perception, and decision-making). Its core goal is to create systems capable of performing tasks that typically require human intelligence.



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AI MAIN CATEGORIES

1. Machine Learning: Allows computers to automatically learn patterns from data without explicit programming. It is the core driving force behind current AI development.
2. Deep Learning: Based on neural networks, it excels at processing complex unstructured data such as images, sound, and text.
3. Natural Language Processing: Enables machines to understand, generate, and respond to human language, such as intelligent dialogue and translation.
4. Computer Vision: Enables machines to "see" images and videos, performing recognition, classification, and tracking.

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AI MAIN CATEGORIES

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- 5. Reinforcement Learning: Allows intelligent agents to learn optimal strategies through interaction with the environment and trial and error; commonly used in game AI and robot control.
- 6. Generative AI: Models capable of creating entirely new content (such as images, music, text, and code), such as AIGC tools.
- 7. Robotics: Combines perception, decision-making, and control to enable robots to perform tasks in the physical world.
- 8. Expert Systems: An early branch of AI that simulates the knowledge and reasoning of human experts in specific domains to make decisions. Artificial Intelligence Ethics and Safety: A cutting-edge interdisciplinary field that studies the fairness, transparency, privacy, and long-term impact of AI.



A FEW YEARS EARLIER...

GARY KASPAROV VS. DEEP BLUE (1997)

- Powered by IBM supercomputer (later also used in Blue Brain Project)
- Deep Blue won the game

Technology

Did Deep Blue beat Kasparov because of a computer bug?

<https://www.wired.co.uk/article/deep-blue-bug>



20 YEARS LATER: GOOGLE DEEPMIND & ALPHAGO

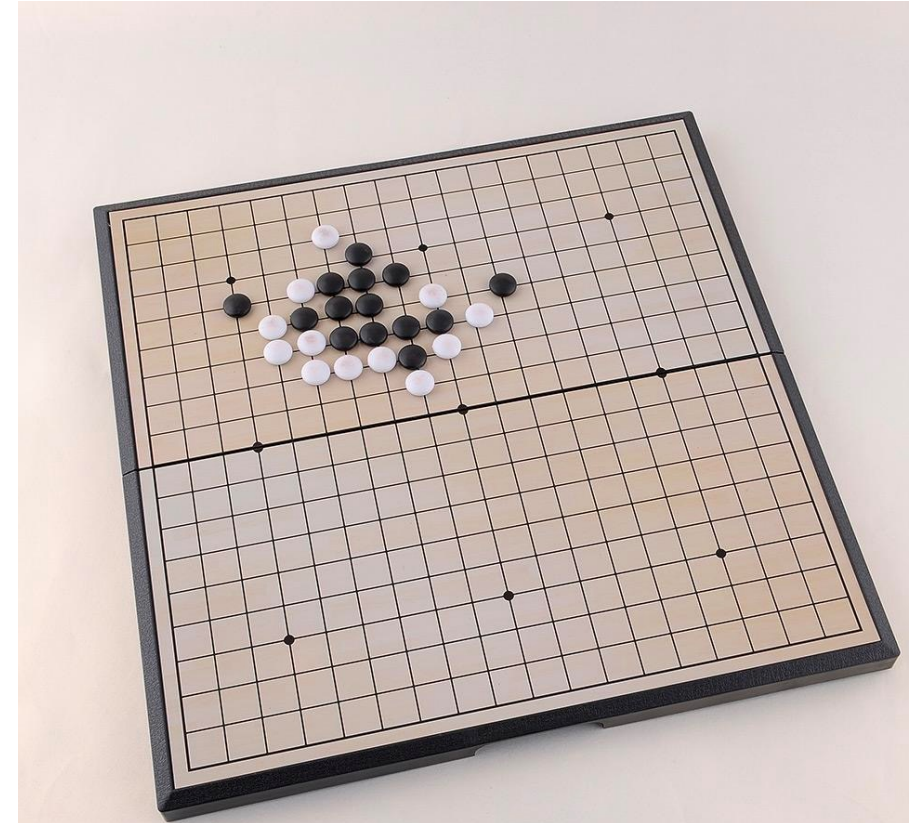
- 19 x 19 board, with way more combinations than in chess
- Search-centered approaches like min-max / alpha-beta inevitably bound to fail
- Evaluation function key. How to know it?
- **Learn it!**
 - Deep reinforcement learning

[ARTICLE](#)

doi:10.1038/nature24270

Mastering the game of Go without human knowledge

David Silver^{1*}, Julian Schrittwieser^{1*}, Karen Simonyan^{1*}, Ioannis Antonoglou¹, Aja Huang¹, Arthur Guez¹, Thomas Hubert¹, Lucas Baker¹, Matthew Lai¹, Adrian Bolton¹, Yutian Chen¹, Timothy Lillicrap¹, Fan Hui¹, Laurent Sifre¹, George van den Driessche¹, Thore Graepel¹ & Demis Hassabis¹



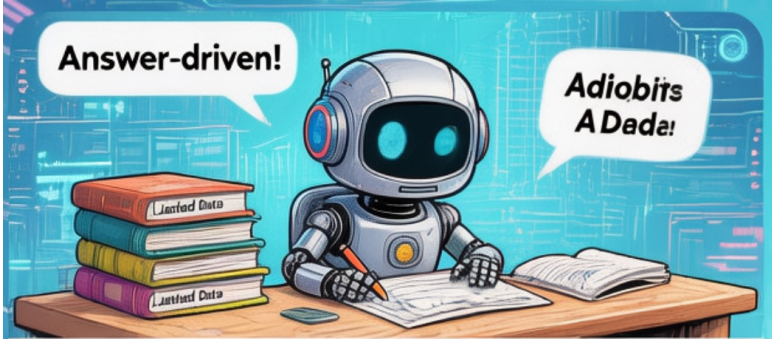
ALPHAGO VS. HUMAN CHAMPIONS

Multiple matches, all ending in AlphaGo victory:

- 2015: 5-0 vs. Fan Hui in Europe
- 2016: 4-1 vs. Lee Sedol in Korea
- 2017: 2-0 vs. Ke Jie in China



WHETHER A LEARNING IS SUPERVISED OR NOT

Learning Paradigm	Core Metaphor	Interesting Core Idea	A Classic/Interesting Application
Supervised Learning	“The Problem Set of a Top Student”	Answer-driven. It believes every problem in the world has a standard answer. Its ultimate goal is to master a huge Standard Answer Handbook and match new questions to the closest answer.	 Answer-driven! Adiobits A Dada! Supervised Learning Label
Unsupervised Learning	“The Explorer’s Map”	Question-driven. It believes the world has no labels—all answers lie hidden in the structure of the data itself. Its mission is to discard preset answers and discover on its own that “these points belong together” or “this axis is the key.”	 Unlabeled Data Unsupervised Learning



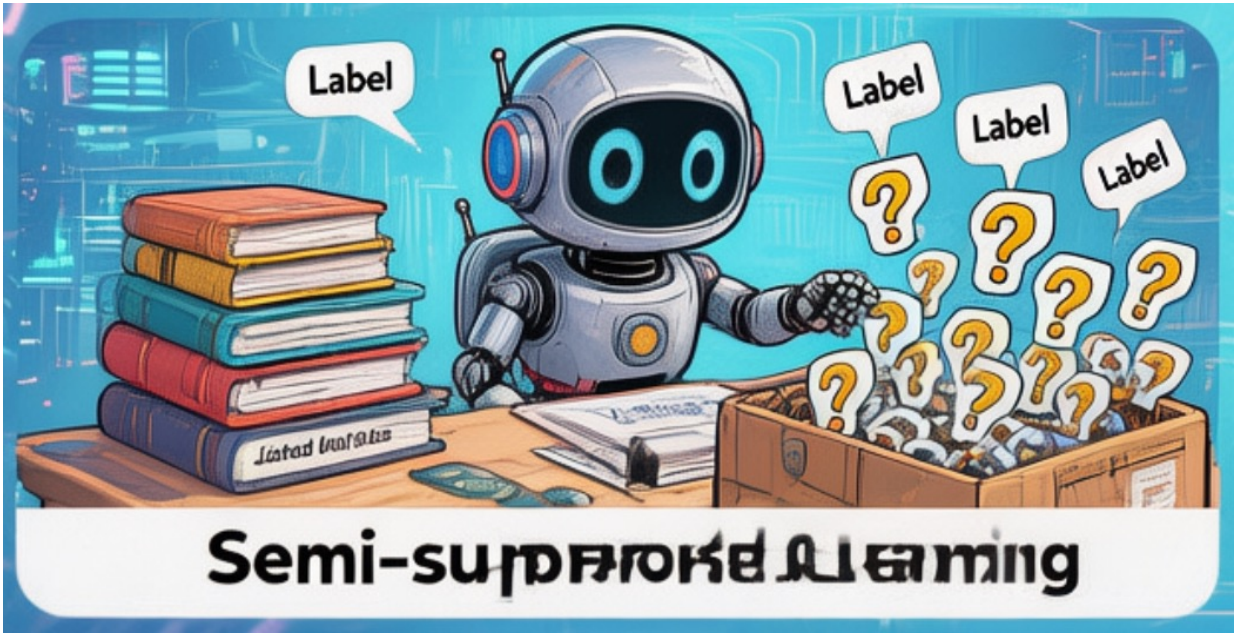
WHETHER A LEARNING IS SUPERVISED OR NOT

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Supervised Learning	“The Problem Set of a Top Student”	Answer-driven. It believes every problem in the world has a standard answer. Its ultimate goal is to master a huge Standard Answer Handbook and match new questions to the closest answer.	Face Recognition: After seeing countless photos of "Zhang San" labeled as "Zhang San", it can recognize him in new photos. It doesn't care why Zhang San looks that way—only about matching.
Unsupervised Learning	“The Explorer’s Map”	Question-driven. It believes the world has no labels—all answers lie hidden in the structure of the data itself. Its mission is to discard preset answers and discover on its own that "these points belong together" or "this axis is the key."	Customer Segmentation: An e-commerce platform doesn't know how many customer types exist, but the algorithm analyzes purchase behavior and automatically groups users into clusters like "Bargain Hunters," "Quality Lifestylers," or "Impulse Shoppers"—often surprising the platform.



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Semi-supervised Learning	“The Teacher’s Assistant”	Leverage Effect. It uses a clever assumption: similar data should have similar labels . A small number of precious labeled examples (the teacher) serve as "seeds" to infect and label vast amounts of unlabeled data (the students), achieving "great impact with little effort."	Webpage Classification: Manually label a few "Sports" and "Finance" webpages, and the algorithm automatically categorizes the entire internet’s pages based on link relationships and content similarity.



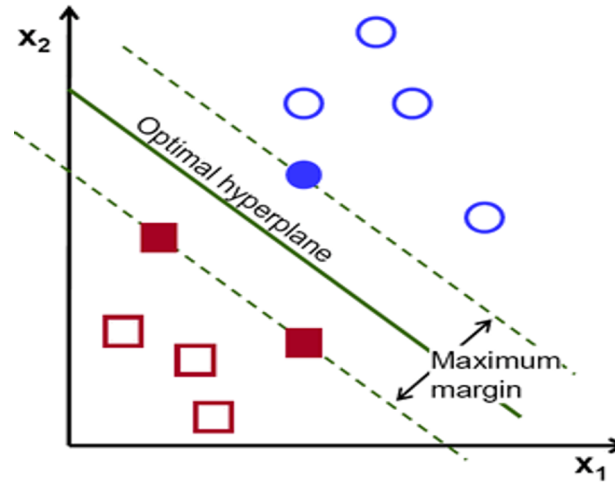
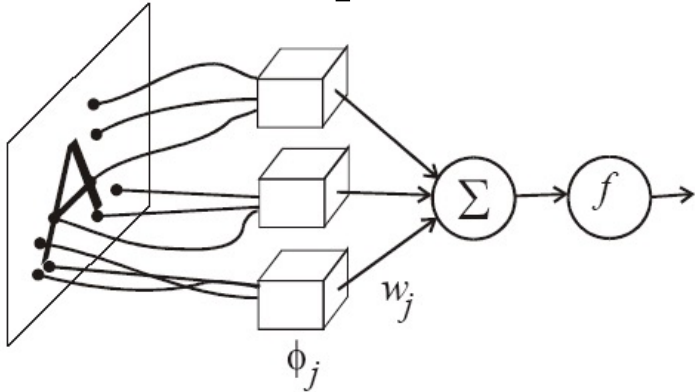
WHETHER A LEARNING IS SUPERVISED OR NOT

Learning Paradigm	Core Metaphor	Interesting Core Idea	A Classic/Interesting Application
Reinforcement Learning	“The Gamer’s Trial and Error”	Consequence-driven. It doesn’t care what the data is, only about the consequences of actions . Through continuous trial and error, it learns to make sequences of decisions in complex environments to maximize long-term reward. Its philosophy: Play the game enough, and you’ll figure out the rules.	AlphaGo: It knows nothing about Go theory or aesthetics—only whether it won (+1 reward) or lost (-1 reward). Through millions of self-play games, it explored strategies that surpassed a millennium of human understanding.

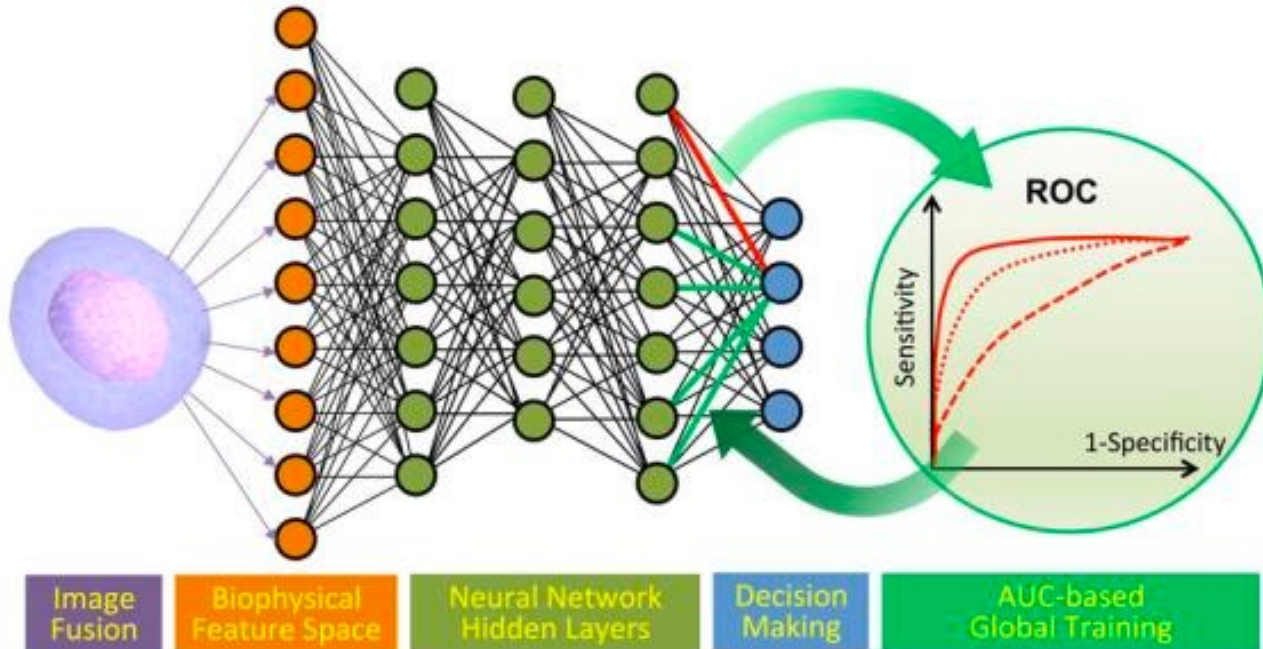
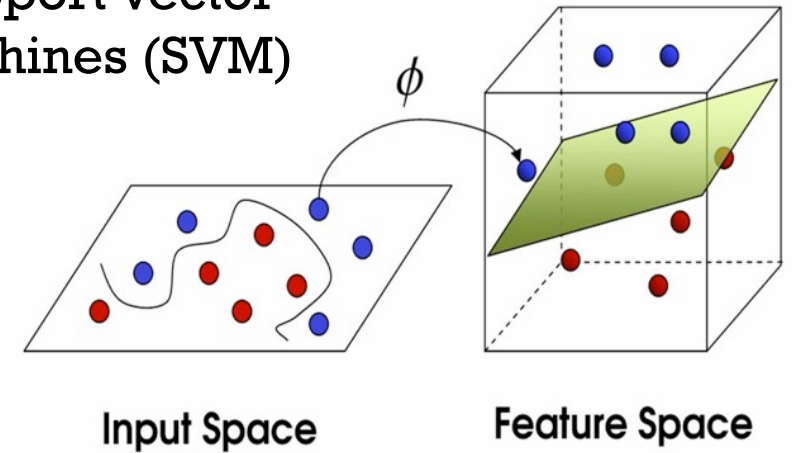


SUPERVISED LEARNING METHODS

Perceptrons

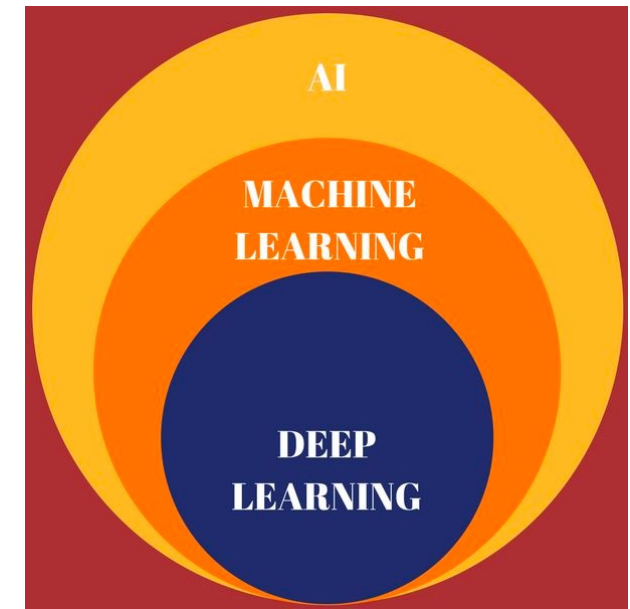


Support vector machines (SVM)



Multi-layer perceptrons (MLP) / Artificial neural networks (ANN)

Deep learning



OBJECT RECOGNITION IN COMPUTER VISION



Based on
large
amounts of
labeled data

Often
acquired
without
(explicit)
consent



IMAGE SEGMENTATION IN COMPUTER VISION

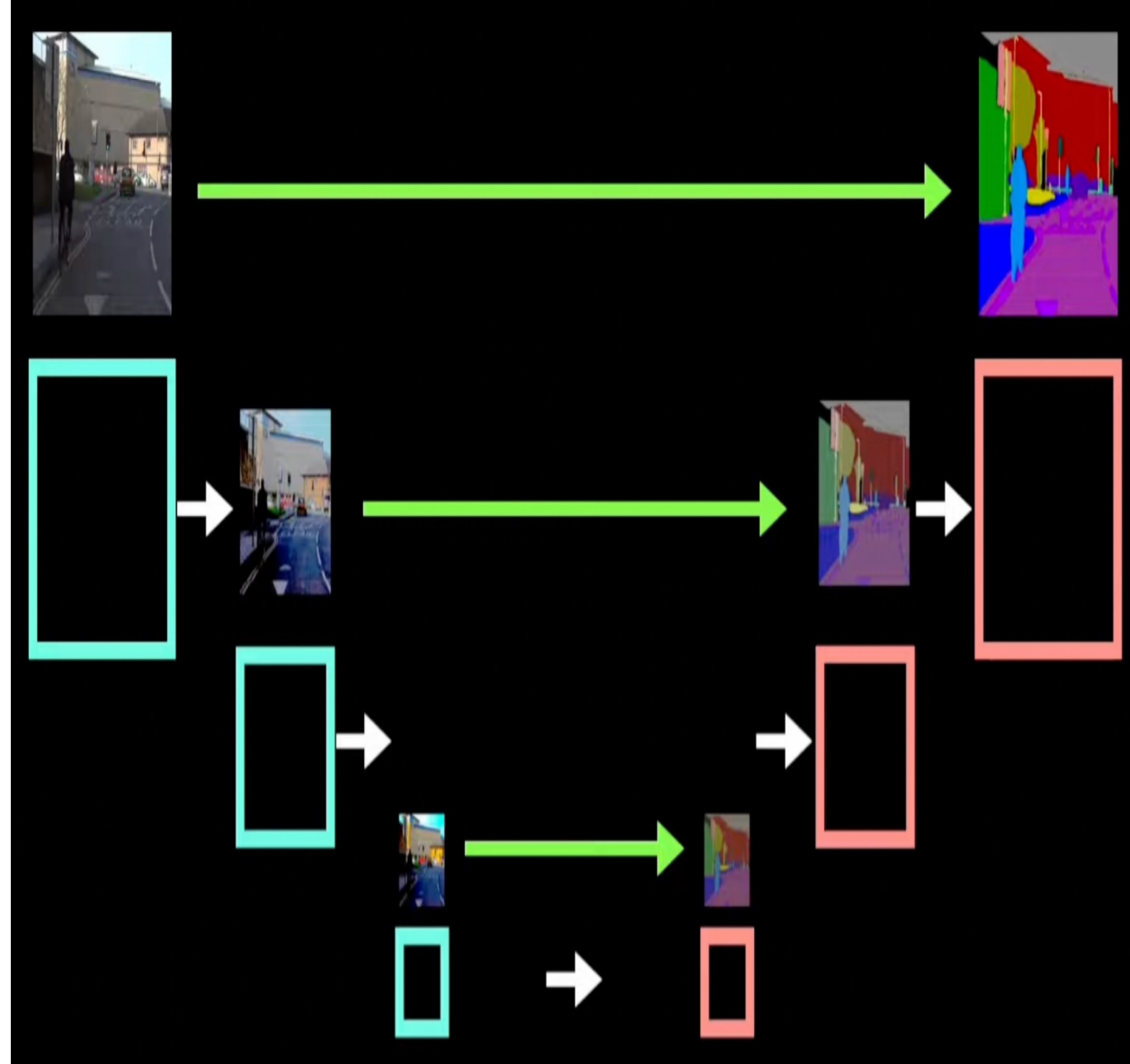
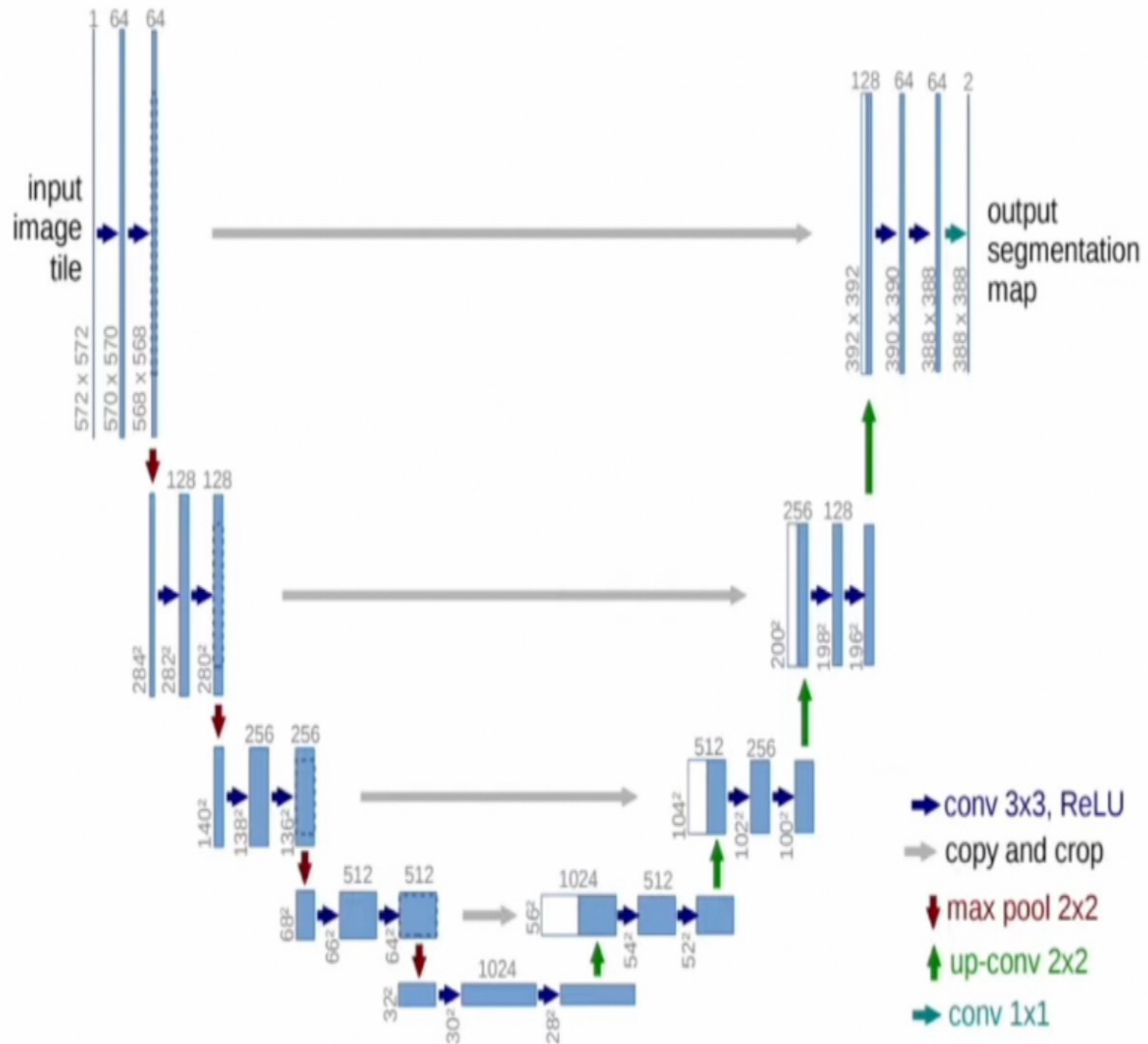


"A scientist holding up a sign that says 'Welcome' on it."

hello and welcome

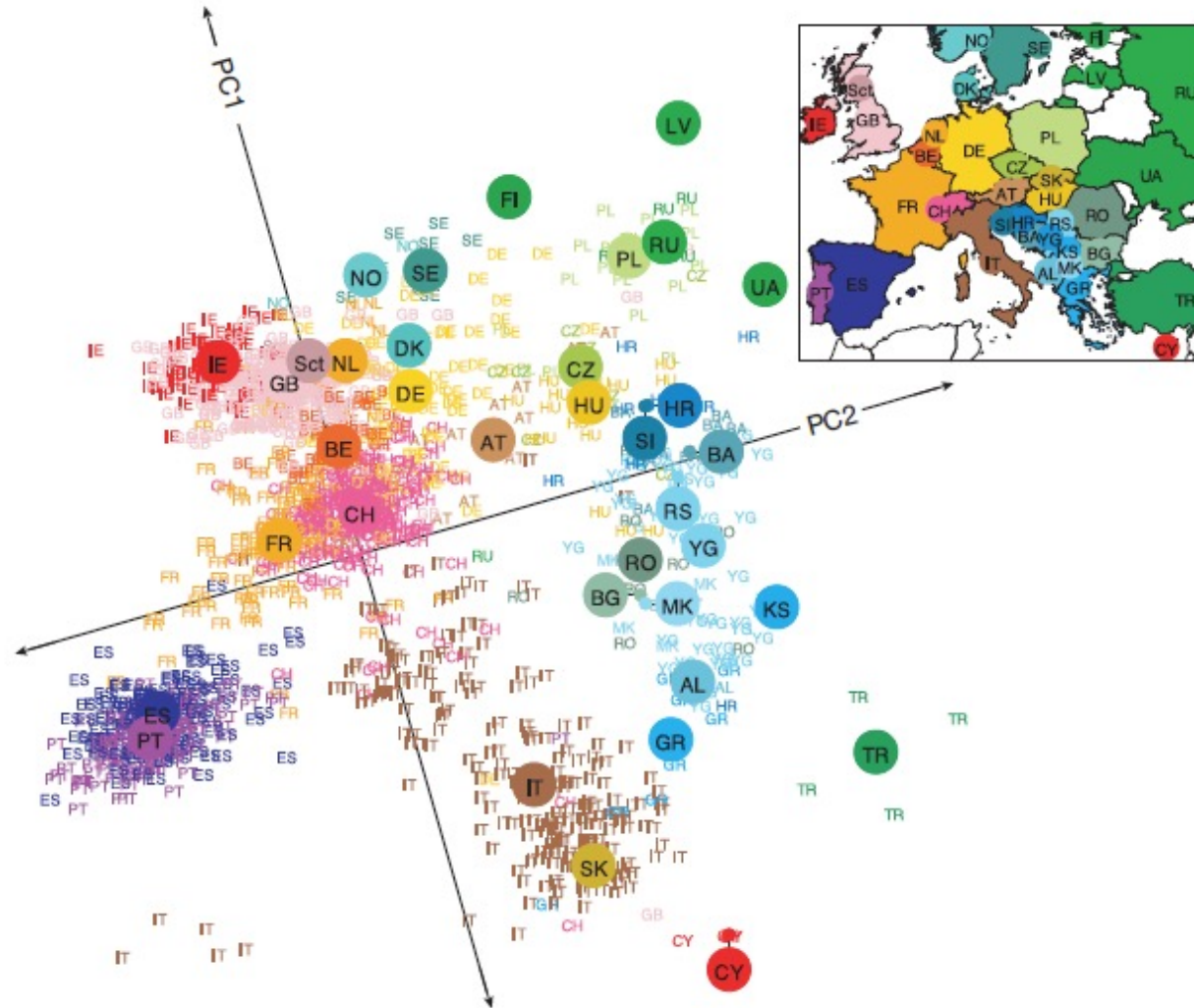


CNN SEGMENTATION: U-NET



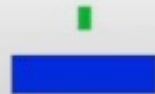
Novembre et al., Nature 2008

- Approximating genetic variation in Europe based on the first 2 principal components
- Predicting one's location based on genes (actually just PCs1 and 2)



LET'S SEE A VERY SIMPLE GAME - PONG

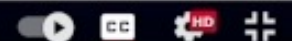
Reinforcement Learning



Steve Seitz



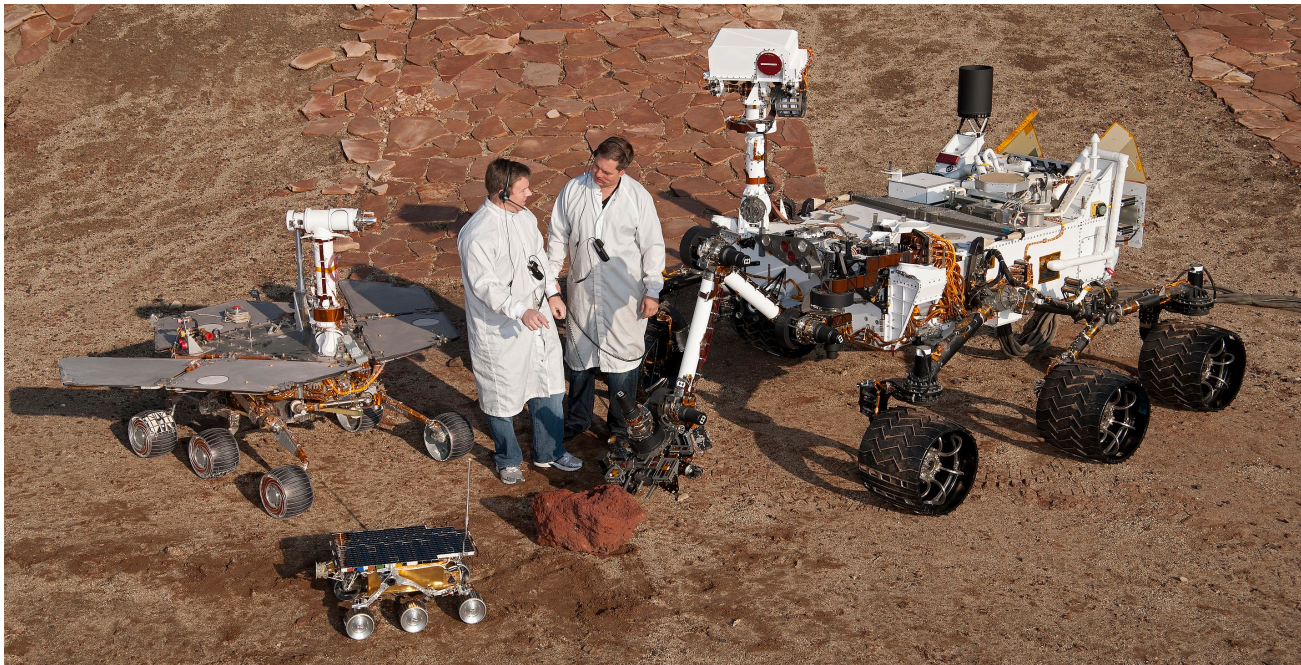
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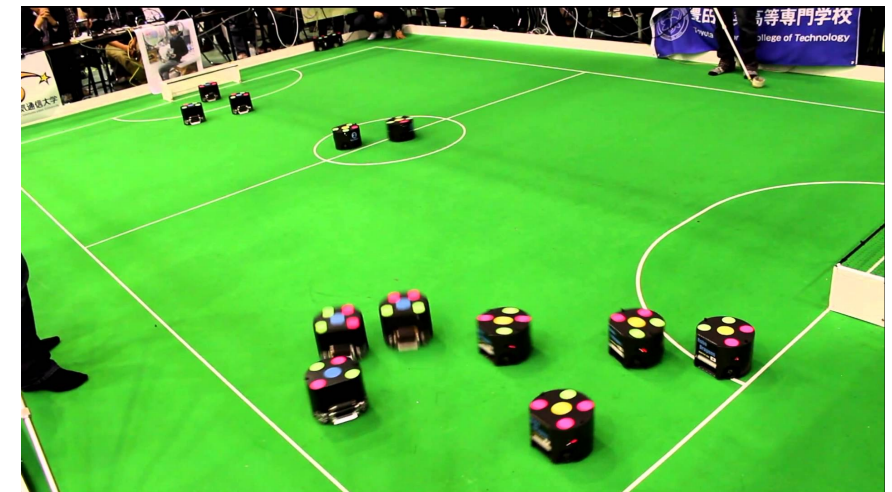
AUTONOMOUS AI: ROBOTICS

Combine multiple AI methods!

Self-driving
yacht



NASA Mars rovers



RoboCup

ROBOTS IN CHINA



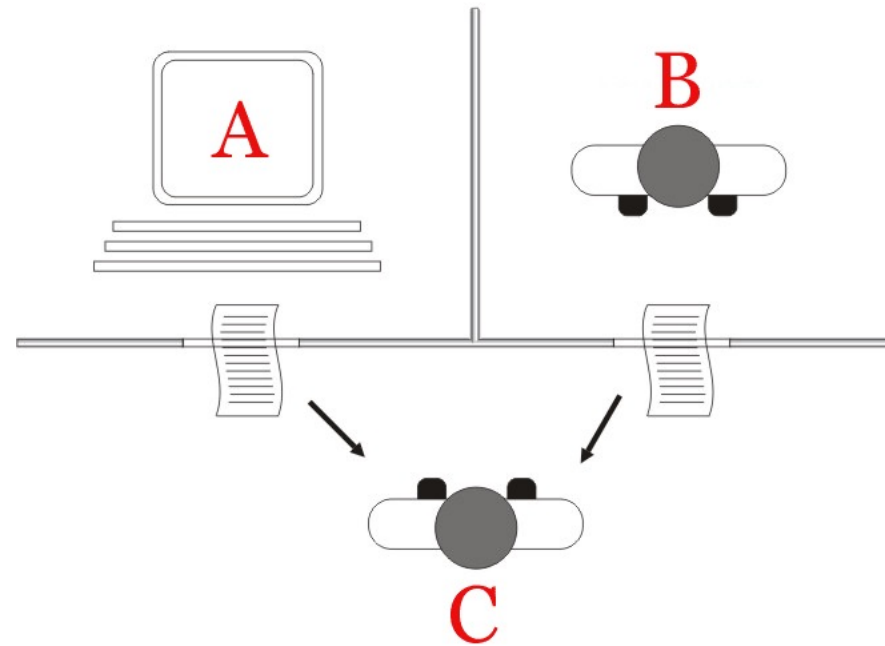
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TRANSITION TO NATURAL LANGUAGE PROCESSING: THE TURING TEST

<https://plato.stanford.edu/entries/turing-test/>

- A test AKA Imitation Game to determine whether a machine can be indistinguishable from humans.
- Requires 3 physically separated terminals: one operated by a computer, two by humans
- One human is a questioner, the other human and computer are respondents



Alan Turing (1912-1954)
Founder of computer science
Breaker of Nazi codes in WW2

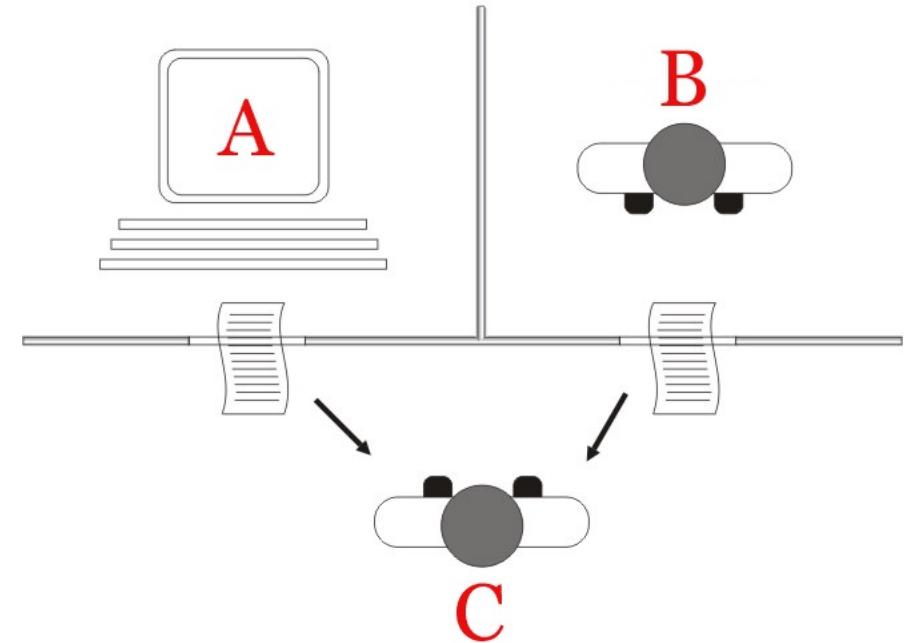
<https://www.turing.org.uk/publications/dnb.html>



TRANSITION TO NATURAL LANGUAGE PROCESSING: THE TURING TEST

<https://plato.stanford.edu/entries/turing-test/>

- The questioner interrogates the respondents using a specified format and context.
- After a certain time the questioner needs to determine which respondent is a computer and which is human
- If after many runs this determination cannot be made, the computer is thought to have artificial intelligence



https://en.wikipedia.org/wiki/Turing_test

**Answer may depend on format,
scope of questioning**

DOES NOWADAYS GPT DISAPPOINT TURING

In a sense, today's large language models like GPT have come very close to, or even partially achieved, Turing's original vision.

However, if judged by the standard of true intelligence (such as understanding, reasoning, and consciousness), current models still fall fundamentally short. They are essentially statistical pattern generators based on massive amounts of data, lacking a true understanding of the world and logical reasoning abilities, and are prone to producing "illusions" (fabricated information).

Therefore, at the level of "behavioral performance," GPT has not disappointed Turing's expectations; but at the level of "the essence of intelligence," it still has a long way to go to reach its vision.



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